

Felix Monninger

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Mathematician (M.Sc., graph theory) and embedded software engineer. Working half-time since June 2026 to build **ArgMap**, a system for making complex arguments explorable and auditable.

EXPERIENCE

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| 2026 – present | Co-organizer , PauseAI München. Co-founded the local AI-safety group; kickoff May 2026 (19 attendees). Venue, recruiting, and event organization (talks, film evenings, outreach). |
| 2021 – present | Embedded Software Engineer , ESR Labs, Munich. Embedded C++/Rust development. Reduced to 50% in June 2026 to self-fund independent work on ArgMap. |
| 2018 – 2021 | Software Engineer , Brainlab AG, Munich. C++ developer in medical technology; medical imaging data processing and network security. |

PROJECTS

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| 2026 | ArgMap , p1graph.org. Built in summer 2026, on mathematical foundations developed over the preceding decade. Interactive argument maps solved as factor graphs: statements carry author-assigned confidences, and an in-browser Rust-to-WASM junction-tree solver reconciles them into a joint distribution in real time, surfacing tensions in the argument. Flagship map: <i>If Anyone Builds It, Everyone Dies</i> (~600 variables, treewidth 11), LLM-extracted from the primary sources under a pre-registered rubric, in four languages. Text-based authoring format (spec frozen at v0.2), eleven readers across nine logged test sessions. Public release on LessWrong: September 2026. |
| 2023 – present | PitchVis , p1graph.org/pitchvis. Real-time musical pitch visualization for choir singers, written in Rust (Bevy). Cross-platform: web (WASM), Android, and a dedicated hardware device; used live in concerts. |

EDUCATION

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| 2013 – 2016 | M.Sc. Mathematics , LMU München, grade 1.66 (1.0 = best). Thesis in graph theory: message-passing algorithms on graphs. Followed by a year of independent study applying methods from statistical physics to factor graphs (2016–17), which became the basis of ArgMap's solver. |
| 2008 – 2012 | B.Sc. Mathematics , TU München. Minor in computer science; specialization in complex analysis and analytic number theory. |
| 2000 – 2008 | Abitur , Karl-Friedrich-Gymnasium Mannheim, grade 1.6. Completed at age 16 (G8); concurrent university coursework in computer science (Universität Mannheim, 2006–07). |

LANGUAGES, SKILLS & VOLUNTEERING

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| Languages | German (native) · English (fluent) · Mandarin (conversationally fluent; self-study and language exchange since 2014) |
| Technical | Rust (WASM, Bevy, probabilistic inference), C++17, Python · graphical models, junction-tree / message-passing algorithms · Linux · LLM-assisted extraction pipelines |
| Volunteering | Treasurer (Finanzvorstand), daycare parents' association, Munich, since 2026 |